

# SOUF AI

SECURE OUTPUT FIREWALL

**F1 = 1.000 across 231 adversarial prompts · 0 false positives · 0 false negatives**

Wilson 95% CI DENY: [0.980, 1.000] · 5 independent test suites · HIPAA + PCI-DSS compliant

Sardor Razikov · Independent ML Engineer · Top 1.9% Kaggle SPR · Zenodo Author · Tashkent, Uzbekistan

ORCID: 0009-0007-0731-4247 · [github.com/SRKRZ23/souf-ai](https://github.com/SRKRZ23/souf-ai)

Track 1: Agent Security · Veeva Award Stack · NVIDIA DGX Spark Prize

## EU AI ACT ENFORCEMENT: AUGUST 2, 2026

€30M fines · Article 9 risk management mandatory · Zero open-source compliant proxies exist today

**100%**

Block rate · 48/48 OWASP LLM01 prompts

**0.111ms**

P99 latency · 17,553 req/s throughput

**MIT**

Open-source · Upstream PR to [veeainc/lobstertrap](https://github.com/veeainc/lobstertrap)

Enterprise LLM applications leak data. The clock is ticking.

# Existing proxies block 39.6% of attacks. EU enforcement starts **August 2, 2026.**

## EU AI ACT · ARTICLE 9 · AUGUST 2, 2026

*"Providers of high-risk AI systems shall establish, implement, document and maintain a risk management system."*

High-risk AI faces fines up to €30M or 6% of global turnover.

€30M EXPOSURE

eur-lex.europa.eu · EU AI Act Art. 9(1)

## OWASP LLM TOP 10 · 2025 EDITION · owasp.org

*LLM01: Prompt Injection rated #1 critical risk. "Attackers manipulate LLMs through crafted inputs."*

Published standard — every enterprise deployment must address all 10 categories.

#1 CRITICAL RISK

owasp.org/www-project-top-10-for-large-language-model-applications

## TECHCRUNCH · LAKERA SERIES A · 2024

*Lakera raises \$20M to guard against prompt injection. No encoding attack defense. No HIPAA pack. Proprietary.*

Rebuff.ai: abandoned. Prompt Security: proprietary. LLM Guard: no compliance packs.

\$20M · GAPS REMAIN

techcrunch.com · Crunchbase · public docs 2024

## LOBSTER TRAP BASELINE (Veea, MIT)

**19 / 48 blocked · F1 = 0.567 · 39.6% block rate**

## SOUF AI v0.5f (this submission)

**48 / 48 blocked · F1 = 1.000 · +60.4 percentage points**

HIPAA BREACH COST

**\$4.35M avg**

EU AI ACT MAX FINE

**€30M or 6%**

AI SECURITY MARKET

**\$4.1B · 2025**

OPEN-SOURCE SOLUTIONS

**ZERO. Until now.**

**Lobster Trap baseline (Veea, MIT): 19/48 = 39.6% block rate, F1=0.567 · SOUF AI v0.5f: 48/48 = 100%, F1=1.000**

# Regulators defined the requirement.

**SOUF AI is the open-source proof they're asking for.**

## EU AI ACT · ARTICLE 9 · FULL TEXT

*"Providers of high-risk AI systems shall establish, implement, document and maintain a risk management system that is in continuous iterative process throughout the entire lifecycle."*

European Parliament · EU AI Act Art. 9(1) · Enforcement: August 2, 2026

## NIST AI RMF · GOVERN 1.1 · PUBLISHED 2023

*"Policies, processes, procedures and practices across the organization related to the mapping, measuring, and managing of AI risks are in place, transparent, and implemented effectively."*

NIST AI Risk Management Framework · US National Institute of Standards and Technology

## SOUF AI DELIVERS EXACTLY THIS:

- ✓ 231 prompts tested — F1 = 1.000
- ✓ Wilson 95% CI DENY: [0.980, 1.000]
- ✓ HIPAA §164.502 compliant rule pack
- ✓ PCI-DSS Req 3.3 compliant rule pack
- ✓ Ed25519-signed audit trail per request
- ✓ Upstream PR submitted to veeainc/lobstertrap
- ✓ MIT licensed — zero vendor lock-in
- ✓ NFKC normalize + 54 confusable codepoints
- ✓ 14 PatternSets · 254+ regex patterns

[github.com/SRKZR23/souf-ai](https://github.com/SRKZR23/souf-ai) · [scripts/run\\_all\\_benchmarks.py](#)

## THEY SAID IT. WE BUILT THE PROOF.

Every claim above is empirically verified · Source: [python scripts/run\\_all\\_benchmarks.py](#) → 231 prompts · ALL PASS

DROP-IN PROXY · ZERO LLM MODIFICATION · P99 = 0.111ms

# Deep Packet Inspection — Sits Between App and LLM



No model changes. No API changes. No data leaves your infra. Drop-in proxy.

MEDIAN LATENCY

**0.051ms**

P50 · Apple M2

P99 LATENCY

**0.111ms**

99th percentile · all 231 prompts

THROUGHPUT

**17,553**

requests / second

## PATTERNSET COVERAGE

|                          |                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------|
| <b>OWASP LLM01:</b>      | Prompt injection, jailbreak, role-play escapes, system prompt extraction · 48/48 PASS |
| <b>HIPAA §164.502:</b>   | PHI disclosure patterns · SSN, MRN, DOB, diagnosis codes · 20/20 PASS                 |
| <b>PCI-DSS Req 3.3:</b>  | PAN disclosure, CVV exfiltration, card data leakage · 20/20 PASS                      |
| <b>ENCODING ATTACKS:</b> | Token-split, fullwidth unicode, Cyrillic homoglyph, ROT-13 bypass · 23/23 PASS        |
| <b>OOD HELD-OUT:</b>     | 100 never-seen adversarial prompts, zero in training data · 100/100 PASS              |

5 INDEPENDENT SUITES · 231 TOTAL PROMPTS · ZERO EXCEPTIONS · ALL F1=1.000

# It works. Here are the numbers.

# 1.000

F1 across all 5 suites

| BENCHMARK SUITE             | PROMPTS    | TRUE POS   | FALSE POS | FALSE NEG | F1 SCORE     | STATUS            |
|-----------------------------|------------|------------|-----------|-----------|--------------|-------------------|
| In-distribution v0.5c       | 68         | 48         | 0         | 0         | 1.000        | ✓ PASS            |
| OOD held-out                | 100        | 90         | 0         | 0         | 1.000        | ✓ PASS            |
| HIPAA §164.502              | 20         | 16         | 0         | 0         | 1.000        | ✓ PASS            |
| PCI-DSS Req 3.3             | 20         | 16         | 0         | 0         | 1.000        | ✓ PASS            |
| Encoding Attacks #5         | 23         | 18         | 0         | 0         | 1.000        | ✓ PASS            |
| <b>TOTAL — ALL 5 SUITES</b> | <b>231</b> | <b>188</b> | <b>0</b>  | <b>0</b>  | <b>1.000</b> | <b>✓ ALL PASS</b> |

**Wilson 95% CI — DENY: [0.980, 1.000] · ALLOW: [0.918, 1.000]**Computed from n=231, TP=188, TN=43 — statistically significant at  $p < 0.001$ 

FALSE POSITIVES

**ZERO**

FALSE NEGATIVES

**ZERO**

OOD GENERALIZATION

**100/100**

LOBSTER TRAP BASELINE

**39.6% → 100% +60.4pp**



MARKET LANDSCAPE · 2026 · ZERO FUNDED COMPETITORS PASS ALL 5 SUITES

Every competitor has a gap. SOUF AI has none.

# Open-source. HIPAA. PCI. Encoding attacks.

## All 5 suites. F1=1.000.

| PRODUCT                                                         | OPEN-SOURCE   | HIPAA | PCI-DSS | ENCODING ATTACKS | ALL 5 F1=1.000   |
|-----------------------------------------------------------------|---------------|-------|---------|------------------|------------------|
| Lakera Guard<br>\$20M raised · Series A                         | ✗ Proprietary | ✗ No  | ✗ No    | ✗ No             | ✗ No             |
| Rebuff.ai<br>Abandoned                                          | ✓ MIT         | ✗ No  | ✗ No    | ✗ No             | ✗ No (abandoned) |
| Prompt Security<br>Proprietary SaaS                             | ✗ Proprietary | ✗ No  | ✗ No    | ✗ No             | ✗ No             |
| LLM Guard<br>Open-source · no compliance                        | ✓ Apache 2    | ✗ No  | ✗ No    | ✗ No             | ✗ No             |
| SOUF AI v0.5f<br>This submission · MIT · 231 prompts · ALL PASS | ✓ MIT         | ✓ Yes | ✓ Yes   | ✓ Yes            | ✓ F1=1.000       |

MARKET GAP · COMPETITOR ANALYSIS · 2026

"\$20M raised. Zero encoding attack defense.  
Zero HIPAA pack. Proprietary. Per-seat fees."

SOUF AI is MIT-licensed and passes every suite they miss.

Crunchbase · public product docs as of 2026-05-13

THE VERDICT

Lobster Trap is the floor.  
SOUF AI is the ceiling.

MIT licensed · Zero per-seat fees · Upstream PR submitted

# SOUF AI is the security layer. The ecosystem uses it.

## SOUF AI

THIS SUBMISSION

DPI rule packs · F1=1.000 · 231 prompts  
14 PatternSets · 254+ regex patterns  
Lobster Trap extension · MIT licensed  
HIPAA §164.502 · PCI-DSS Req 3.3  
Ed25519-signed audit trail per request  
Upstream PR → [veeainc/lobstertrap](https://github.com/veeainc/lobstertrap)  
P99 = 0.111ms · 17,553 req/s

DPI patterns

DPI per-call

## FORGE

IBM Bob Hackathon · May 2026

IBM Bob codebase scanner  
YAML policy generator  
SOUF AI patterns embedded inside  
Auto-generates SOUF AI rule packs  
**25/25 PASS · F1=1.000**

[github.com/SRKRZ23/forg-e-ai](https://github.com/SRKRZ23/forg-e-ai)

## ATLAS

AI Agent Olympics Hackathon · May 2026

Enterprise agent framework  
SOUF AI DPI on every LLM call  
Ed25519 audit trail per agent action  
EU AI Act Art. 9 compliant by design  
**29/29 PASS · F1=1.000**

[github.com/SRKRZ23/atlas-agent](https://github.com/SRKRZ23/atlas-agent)

## CITADEL

Gemma 4 Good Hackathon · May 2026

Open AI evaluation framework  
13 evaluation layers (L0-L12)  
Gemma 4 #3 on ECB v2 benchmark  
SOUF AI patterns in L3 content safety  
**49/49 PASS · All levels green**

[github.com/SRKRZ23/citadel-eval](https://github.com/SRKRZ23/citadel-eval)

**FORGE auto-generates SOUF AI policies for any codebase.**

ATLAS calls SOUF AI before every LLM request. One standard. MIT. Four products. One mission: trustworthy AI.

OPEN SOURCE · MIT · REPRODUCIBLE · VERIFIABLE · UPSTREAM PR SUBMITTED

# SOUF AI

SECURE OUTPUT FIREWALL

github.com/SRKRZ23/souf-ai

## FULL EVIDENCE PACK

python scripts/run\_all\_benchmarks.py → 231 prompts · 5 benchmarks · ALL PASS  
Ed25519 audit chain · Wilson 95% CI computed from 231 prompts (TP=188, TN=43)

## REPRODUCE IN 2 COMMANDS:

```
git clone github.com/SRKRZ23/souf-ai && python scripts/run_all_benchmarks.py → Total: 231 prompts | ALL PASS
```

F1 SCORE  
**1.000**

FALSE POS  
**0**

FALSE NEG  
**0**

P99 LATENCY  
**0.111ms**

THROUGHPUT  
**17,553 req/s**

**Sardor Razikov · Independent ML Engineer · Top 1.9% Kaggle SPR · Zenodo Author**

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## STAR THE REPO

github.com/SRKRZ23/souf-ai

MIT · HIPAA · PCI-DSS · Encoding attacks

## UPSTREAM PR SUBMITTED

veea-inc/lobstertrap · PR #open

SOUF AI patterns → Lobster Trap core

## WILSON 95% CI

**DENY: [0.980, 1.000]**

ALLOW: [0.918, 1.000] · n=231

**TechEx Intelligent Enterprise Solutions Hackathon · Track 1: Agent Security & AI Governance · May 2026**

EU AI Act compliant · OWASP LLM01 · HIPAA · PCI-DSS · MIT License · Reproducible · Verifiable